

## SEMESTER S6

### DIGITAL IMAGE PROCESSING

(Common to CS/CM/CA/AM)

|                                           |                 |                    |                |
|-------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                        | <b>PECST636</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L:T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                            | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>             | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To provide foundational concepts of digital image representation, processing, and analysis, including image digitization, color theory, and various data structures, to effectively manipulate and analyze digital images.
2. To help the learner develop the ability to implement advanced image processing techniques, such as image segmentation, edge detection, and image compression, while critically evaluating the performance and quality of these methods in practical applications.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Contact Hours</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | The image, its representation and properties - Image representations, Image digitization, Sampling, Quantization, Digital image properties, Metric and topological properties of digital images, Histograms, Entropy, Visual perception of the image, Image quality, Noise in images; Color images - Physics of color, Color perceived by humans, Color spaces, Color constancy; Data structures for image analysis - Levels of image data representation, Traditional image data structures - matrices, Chains, Topological data structures - Relational structures, Hierarchical Data Structures, Pyramids, Quadrees, Other pyramidal structures. | <b>9</b>             |
| <b>2</b>          | Image pre-processing - Pixel brightness transformations-, Position-dependent brightness correction, Gray-scale transformation, Geometric Transformations - Pixel coordinate transformations, Brightness interpolation.<br>Local pre-processing, Image Smoothing, Edge detectors, Zero-crossings the second derivative, Scale in Image Processing, Canny Edge Detection,                                                                                                                                                                                                                                                                             | <b>8</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          | Parametric Edge Models, Edges Multi-spectral images,, Line detection by local pre-processing operators, Detection of corners(interest points), Image Restoration - Degradations that are easy to restore, Inverse Filtering, Wiener Filtering                                                                                                                                                                                                                                                                                                                                                          |           |
| <b>3</b> | Image Segmentation - Thresholding, Threshold Detection Methods- Optimal thresholding, Multi-spectral thresholding, Edge-based segmentation, Edge Image Thresholding, Edge Relaxation, Border Tracing, Border Detection As Graph Searching, Border Detection As Dynamic Programming, Hough Transforms, Border Detection Using Border location information, Region construction from borders, Region-based segmentation - Region merging, Region Splitting - Splitting And Merging, Watershed segmentation Matching, Template Matching, Control Strategies Templating, Evaluation Issues In Segmentation | <b>9</b>  |
| <b>4</b> | Image Transforms - Discrete Cosine Transform, Wavelet transform, Eigen-analysis, Singular value decomposition, Principal component analysis Radon Transform;<br>Image Compression - Image data Properties, Discrete Image Transforms In Image data compression, Predictive compression methods, Vector quantization, Hierarchical and Progressive Compression methods, Comparison Of Compression Methods, JPEG and MPEG image compression JPEG still image compression, JPEG–2000 compression, MPEG full-motion video compression.                                                                     | <b>10</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

### End Semester Examination Marks (ESE)

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                              | Part B                                                                                                                                                                                                                                                                            | Total     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p><b>(8x3 =24 Marks)</b></p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 subdivisions.</li></ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                      | Bloom's Knowledge Level (KL) |
|----------------|------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Understand the properties of monochrome and colour images and the data structures for image analysis | <b>K2</b>                    |
| <b>CO2</b>     | Apply different preprocessing techniques to visualize image enhancement                              | <b>K3</b>                    |
| <b>CO3</b>     | Understand the concept of image segmentation and various techniques used for this.                   | <b>K2</b>                    |
| <b>CO4</b>     | Understand the various transforms used for image processing                                          | <b>K2</b>                    |
| <b>CO5</b>     | Understand the concept of image compression and apply various image compression techniques.          | <b>K2</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 2    |
| <b>CO2</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 2    |
| <b>CO3</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 2    |
| <b>CO4</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 2    |
| <b>CO5</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                               |                                         |                              |                         |
|-------------------|-----------------------------------------------|-----------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                      | <b>Name of the Author/s</b>             | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Image Processing, Analysis and Machine Vision | Milan Sonka, Vaclav Hlavac, Roger Boyle | Cengage                      | 4/e, 2015               |

| <b>Reference Books</b> |                                         |                                          |                              |                         |
|------------------------|-----------------------------------------|------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                | <b>Name of the Author/s</b>              | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Fundamental of Digital Image Processing | Anil K. Jain                             | Pearson                      | 1/e, 2015               |
| 2                      | Digital image Processing                | Ralph Gonzalez, Richard Woods            | Pearson                      | 4/e, 2018               |
| 3                      | Digital Image Processing                | S Jayaraman, S Esakkirajan, T Veerakumar | McGraw Hill                  | 2/e, 2020               |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                              |
| <b>1</b>                              | <a href="https://archive.nptel.ac.in/courses/117/105/117105135/">https://archive.nptel.ac.in/courses/117/105/117105135/</a> |
| <b>2</b>                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105032/">https://archive.nptel.ac.in/courses/106/105/106105032/</a> |